

Short Report:

- **Introduction**

In this task, I analyzed an ARP packet capture using Python and Scapy. The goal was to understand how devices on the network are mapping IP addresses to MAC addresses and to check whether anything unusual or suspicious is happening, like possible ARP spoofing.

- **What I did**

I loaded the PCAP file using Scapy and filtered only ARP packets. Then I extracted the sender IP addresses and their MAC addresses and built a simple mapping table to see how devices are communicating on the network.

- **What I found**

Most of the network looked normal. Most IP addresses were mapped to a single MAC address, which is what you'd expect in a normal LAN.

However, one thing stood out:

-The IP address 192.168.1.1 was linked to two different MAC addresses:

- 66:66:66:66:66:66
- 00:aa:bb:cc:dd:01

This is not usually normal because one IP should normally belong to one device.

I also noticed some gratuitous ARP packets, where devices were announcing their own IP-to-MAC mappings without being asked. These were seen from:

- 192.168.1.50
- 192.168.1.30
- 192.168.1.1

- **What this might mean**

The duplicate MAC addresses for 192.168.1.1 could be a sign of ARP spoofing or ARP poisoning, where a device is trying to pretend it owns that IP address. This can be used to intercept or redirect network traffic.

The gratuitous ARP packets are not necessarily bad on their own, but combined with the duplicate mapping, they make the situation worth looking into more closely.

- **Conclusion**

Overall, most of the network looked normal and the IP-to-MAC mappings were what you'd expect. But one thing that stood out was that 192.168.1.1 showed up with two different MAC addresses, which is not really normal and could mean something weird is going on, like ARP spoofing or a misconfigured device. The gratuitous ARP messages also showed up a few times, which can be normal, but combined with the duplicate mapping, it's worth checking that device just to b

